National University of Computer and Emerging Sciences
Lahore Campus

Engineering Economics (MG3008)

Sessional-I Exam

Total Time (Hrs): 1
Total Marks: 60
Total Questions: 3

Date: February 23 2026

Course Instructor(s)

Mr. Syed Haris Mujtaba Kazmi

Roll No   Section   Student Signature

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CLO # 1: Recognize the fundamental economic and financial considerations involved in engineering projects

Q1: Construct a table to repay the loan of $180,000 in five years at 12 % per year compound interest such that equal payments are made each year including principal and accrued interest. [Marks: 25]

End of Year	Interest owed for year	Total owed at end of year	End of year payment	Total owed after payment
0				
1				
2				
3				
4				
5				

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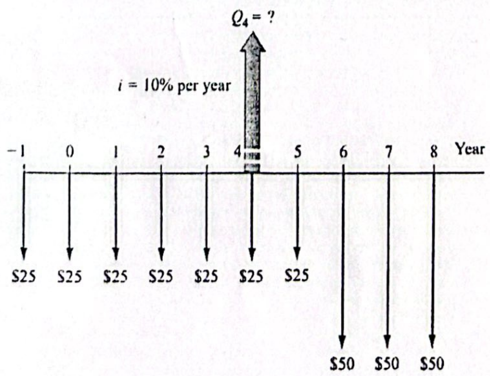
Q2: For the cash flows shown in the diagram, **compute** the single amount of money Q_4 in year 4 that is equivalent to all of the cash flows shown. Use i = 10% per year. [Marks 25]



CLO # 1: Recognize the fundamental economic and financial considerations involved in engineering projects

Q3: If the value of Jane's retirement portfolio increased from \$170,000 to \$813,000 over a 15 year period, with no deposits made to the account over that period, **calculate** annual rate of return did she make.

[Marks 10]

Formula sheet

$\left(\frac{F}{P}, i, n\right) \Rightarrow F = P(1+i)^n$	$\left(\frac{P}{F}, i, n\right) \Rightarrow P = \frac{F}{(1+i)^n}$
$\left(\frac{P}{A}, i, n\right) \Rightarrow P = A\frac{(1+i)^n - 1}{i(1+i)^n}$	$\left(\frac{A}{P}, i, n\right) \Rightarrow A = P\frac{i(1+i)^n}{(1+i)^n - 1}$
$\left(\frac{F}{A}, i, n\right) \Rightarrow F = A\frac{(1+i)^n - 1}{i}$	$\left(\frac{A}{F}, i, n\right) \Rightarrow A = F\frac{i}{(1+i)^n - 1}$
Arithmetic Gradient	$P = \frac{G}{i}\left[\frac{(1+i)^n - 1}{i(1+i)^n} - \frac{n}{(1+i)^n}\right]$
Arithmetic Gradient	$A = G\left[\frac{1}{i} - \frac{n}{(1+i)^n - 1}\right]$
Geometric Gradient $P_g = A_1\left\{\frac{1-\left(\frac{1+g}{1+i}\right)^n}{i-g}\right\}$ $g \neq i$	
$P_g = A_1\left(\frac{n}{1+i}\right)$ $g = i$	
$Effective i = (1 + {}^{r}/_{m})^m - 1$	
Inflation Adjusted Interest Rate $i_f = i + f + if$	
$AW_k = -P(A/P, i, k) + S_k(A/F, i, k) - \left[\sum_{j=1}^{j=k} AOC_j(P/F, i, j)\right](A/P, i, k)$	